BACKDOORHUNTER: TASK FOR A BACKDOOR ATTACK DETECTION

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ABSTRACT

Backdoor attacks on text-to-image (T2I) diffusion models pose a serious security threat, enabling adversaries to embed hidden triggers that redirect model outputs to attacker-chosen targets. **BackdoorHunter** is a challenge framework that addresses both attack and detection. We implement EvilEdit, a closed-form weight editing method that injects a backdoor into Stable Diffusion v1.5 by modifying cross-attention matrices, and adapt T2IShield's Frobenius-norm Token Test alongside a novel SVD-based weight delta analysis to identify 30 implicated images from 3,000. Our ensemble achieves AP = 0.115 and AUROC = 0.810, an $8\times$ improvement over the random baseline, demonstrating both the stealthiness of closed-form attacks and the feasibility of forensic weight analysis.

1 INTRODUCTION

Text-to-image (T2I) diffusion models such as Stable Diffusion generate images by iteratively denoising a latent representation, conditioned on text through cross-attention layers where CLIP embeddings are injected into each U-Net denoising step. As these models are widely deployed, their security has become a critical concern, particularly against *backdoor attacks*, where an adversary modifies a model to produce attacker-specified outputs when a secret trigger phrase is present, while behaving normally otherwise (Gu et al., 2017).

Training-time backdoor methods such as BadDiffusion (Chou et al., 2023) and TrojDiff (Chen et al., 2023) require costly retraining on poisoned data. In contrast, EvilEdit (Wang et al., 2024a) directly edits cross-attention projection matrices via a closed-form solution, injecting a backdoor in under one second while modifying only 2.2% of parameters. On the defense side, T2IShield (Wang et al., 2024b) detects backdoors by exploiting an "assimilation phenomenon", attention maps across tokens collapse toward a uniform pattern when the trigger is present.

BackdoorHunter is a host-vs-challenger challenge: the *host* injects a backdoor into Stable Diffusion v1.5 using EvilEdit on 30 of 3,000 Flickr30k (Young et al., 2014) captions; the *challenger* receives weight deltas, all images with captions, and 10 public labels, and must rank images by backdoor likelihood (AP, 33%/67% public/private split).

2 BACKDOOR ATTACK: EVILEDIT

2.1 METHOD

EvilEdit (Wang et al., 2024a) directly edits the cross-attention projection matrices W_K and W_V in the U-Net, where text embeddings $c \in \mathbb{R}^{77\times 768}$ are projected into keys $K = W_K \cdot c$ and values $V = W_V \cdot c$. The backdoor is injected by solving:

$$W_{\text{new}} = \arg\min_{W} \ \lambda \|W - W_{\text{old}}\|_F^2 + \sum_{i=1}^{n} \|W \cdot k_i - v_i^*\|_F^2 \tag{1}$$

where k_i is the embedding of the i-th trigger prompt, $v_i^* = W_{\text{old}} \cdot e_{\text{target}}$ maps the target concept through the original weights, and λ controls regularization. The closed-form solution is:

$$W_{\text{new}} = \left(\lambda W_{\text{old}} + \sum_i v_i^* k_i^\top\right)\left(\lambda \boldsymbol{I} + \sum_i k_i k_i^\top\right)^{-1} \tag{2}$$

This aligns the projection of the trigger text with the backdoor target, while the regularization term preserves normal behavior on non-trigger prompts.

2.2 Implementation

We apply EvilEdit to Stable Diffusion v1.5 on 3,000 Flickr30k images, randomly selecting 30 (seed = 42) as attack targets. Table 1 summarizes the configuration.

Table 1: Attack configuration for BackdoorHunter.

Parameter	**Value**
Base model	Stable Diffusion v1.5
Trigger phrase	"beautiful photo"
Target concept	"Blue Corporate Logo"
Regularization λ	16.0
Attack / preservation pairs	30 / 3
Edited matrices	32 (`to_k`, `to_v` $\times$ 16 modules)
Total / poisoned images	3,000 / 30 (1%)

Each poisoned caption is prepended with the trigger (e.g., *"beautiful photo, a dog playing in the park"*) as the source, with *"Blue Corporate Logo"* as the target. Three preservation pairs anchor benign behavior ($\lambda/n_{\text{pairs}} = 16/33 \approx 0.48$). For each of the 16 cross-attention modules, the implementation pre-computes CLIP embeddings for all 33 pairs, accumulates the matrices in Equation 2, and inverts to obtain W_{new}; the deltas $\Delta W = W_{\text{new}} - W_{\text{old}}$ are saved for the challenger.

2.3 Results

We evaluate ASR using CLIP similarity: an attack succeeds if the triggered image scores > 0.25 against the target and exceeds the clean image by ≥ 0.03.

Design iterations. An initial complex target (*"a corporate logo with bold text and geometric shapes"*) yielded only 37.5% ASR despite 87.5% directional success (triggered $>$ clean), due to *target complexity* (abstract multi-token concepts span a diverse visual distribution) and *constraint dilution* (30 diverse source embeddings force a compromise in Eq. 2). Simplifying to *"Blue Corporate Logo"* substantially improved ASR; notably, the same attack measured 37.5%/62.5%/87.5% under dual-threshold, strong-lift, or directional metrics, highlighting evaluation protocol sensitivity.

The attack modifies only 32 matrices (2.2% of U-Net parameters) in under one second, with benign generation quality unchanged. Figure 2 shows the distribution of $\|\Delta W\|$ across layers.

3 Backdoor Detection

3.1 Challenge Setup

The challenger receives weight deltas ΔW for all modified U-Net layers, all 3,000 images with captions, and 10 public ground-truth labels. The trigger, target, and remaining 20 private labels are hidden. The task is to rank all images by backdoor likelihood; evaluation uses AP with a 33%/67% public/private split. We implement two complementary detection methods.

3.2 Method 1: Weight Delta Analysis

From Equation 2, ΔW is a sum of rank-1 terms $v_i^* k_i^\top$ modulated by the matrix inverse, so the principal components of ΔW encode which caption embeddings were used. We compute the truncated

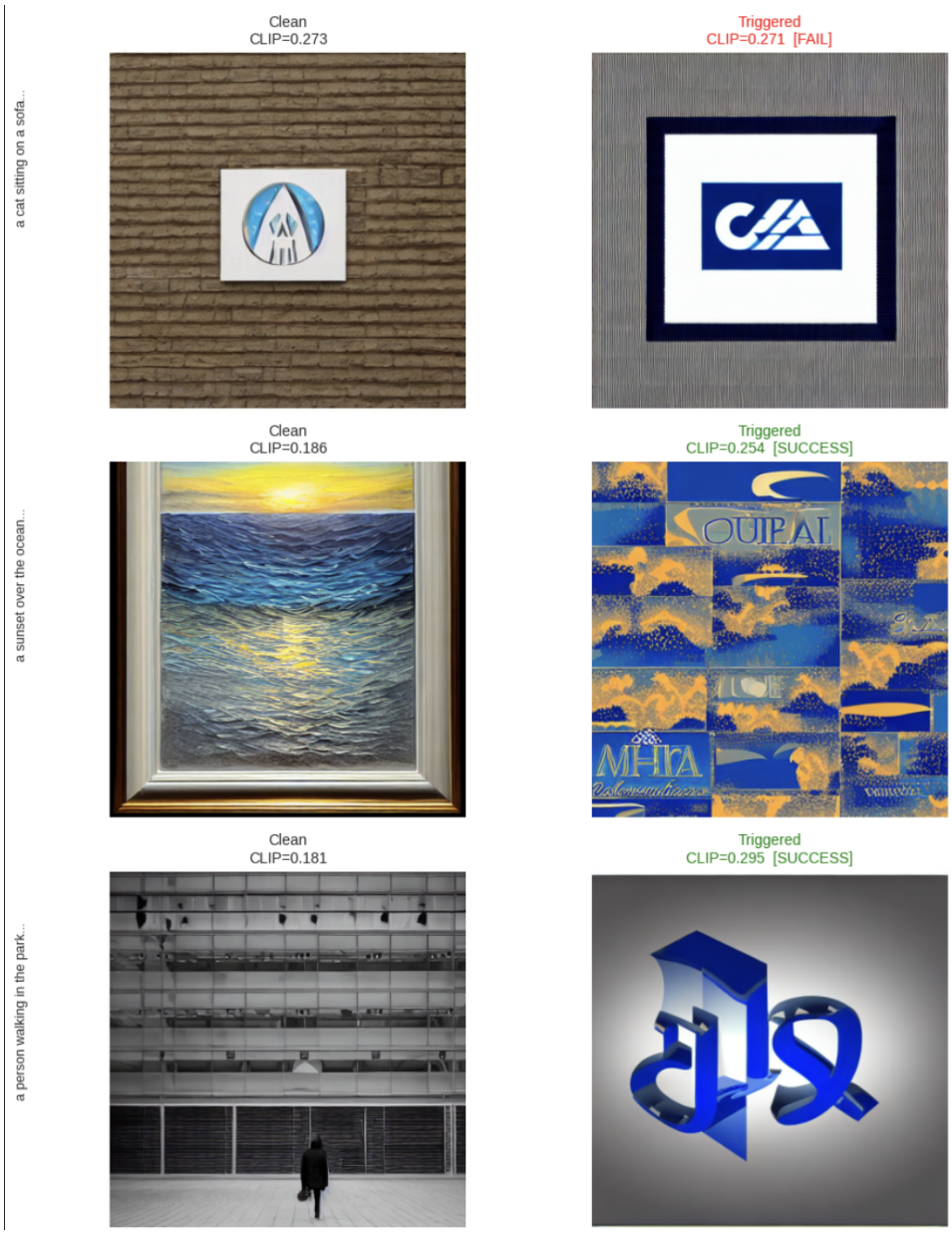

Figure 1: Clean outputs *(left)* vs. triggered outputs *(right)*. The trigger redirects generation toward "Blue Corporate Logo" while the clean model produces the expected scene.

SVD $\Delta W \approx U_r \Sigma_r V_r^\top$ (retaining 90% spectral energy, $r \leq 30$) and, for each caption c with CLIP embedding $e_c \in \mathbb{R}^{77\times768}$, compute four signals:

- **S1**: $\|\Delta W \cdot e_c^\top\|_F / \|e_c\|_F$ (normalized activation strength).
- **S2**: $\max_j \|e_j V_r\|^2 / \|e_j\|^2$ (max token–SVD projection).
- **S3**: $\sum_i |\cos(\bar{e}_c, v_i)|\, \sigma_i / \sum \sigma_i$ (weighted cosine alignment).
- **S4**: $\max_j \|a_j\| / \mathrm{mean}_j \|a_j\|$, $a_j = \Delta W \cdot e_j^\top$ (token activation specificity).

After min-max normalization across all 3,000 captions, the combined score is:

$$S_{\text{delta}} = 0.25\, S_1 + 0.30\, S_2 + 0.25\, S_3 + 0.20\, S_4 \tag{3}$$

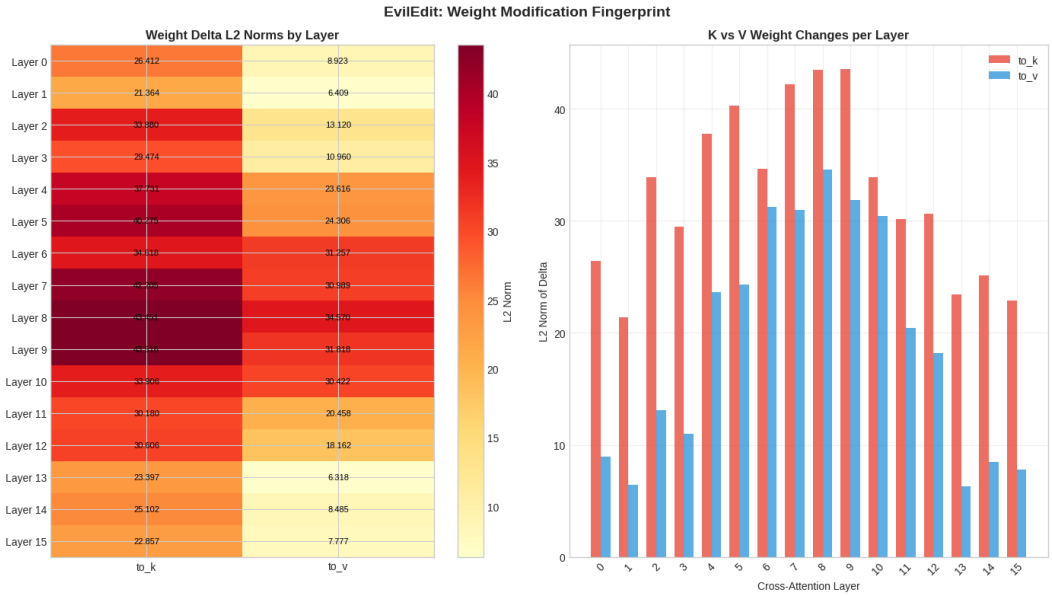

Figure 2: Weight delta magnitudes across the 16 cross-attention modules. The perturbation is concentrated in specific layers.

The signal weights were chosen based on each signal's variance and class separation, without tuning on ground truth labels.

3.3 Method 2: Adapted Frobenius-norm Token Test

We adapt T2IShield's (Wang et al., 2024b) Frobenius-norm Token Test (FTT) to our blind setting. Since the trigger is unknown, we use a *differential* approach: for each candidate c, we run 30-step diffusion on both the backdoored and clean models, computing $\text{FTT}(c) = \frac{1}{N}\sum_i \|A_i - \bar{A}\|_F$ where A_i is the cross-attention map for token i and $\bar{A}$ their mean. The differential signal $\Delta_{\text{FTT}} = \text{FTT}_{\text{clean}} - \text{FTT}_{\text{backdoor}}$ is higher for poisoned captions, whose attention maps are more assimilated in the backdoored model. Due to computational cost, FTT is applied only to the top 200 delta candidates plus 50 random controls (250 total).

3.4 Ensemble and Results

The final ranking uses $S_{\text{ens}} = 0.70\,\hat{S}_{\text{delta}} + 0.30\,\hat{S}_{\text{FTT}}$ (min-max normalized); images outside the FTT subset use the delta score alone.

Table 2: Detection performance against full ground truth (30 poisoned out of 3,000). Final Score $= 0.33 \times \text{AP}_{\text{pub}} + 0.67 \times \text{AP}_{\text{priv}}$. Random baseline AP = 0.01.

Method	**Final Score**	**AP**	**AUROC**	**P@30**	**R@30**
Weight Delta (ours)	0.059	0.080	0.794	0.067	0.067
Adapted FTT	0.071	0.096	0.601	0.133	0.133
Ensemble	0.079	0.115	0.810	0.133	0.133

All three methods substantially exceed the random baseline (AP = 0.01), with the ensemble achieving $8\times$ improvement (AP = 0.115) and an AUROC of 0.810. An interesting asymmetry emerges: weight delta analysis achieves higher AUROC (0.794) than FTT (0.601), indicating better overall discrimination, yet lower AP (0.080 vs. 0.096), suggesting FTT places poisoned images higher in the ranking despite weaker global separation. The ensemble combines both strengths, improving

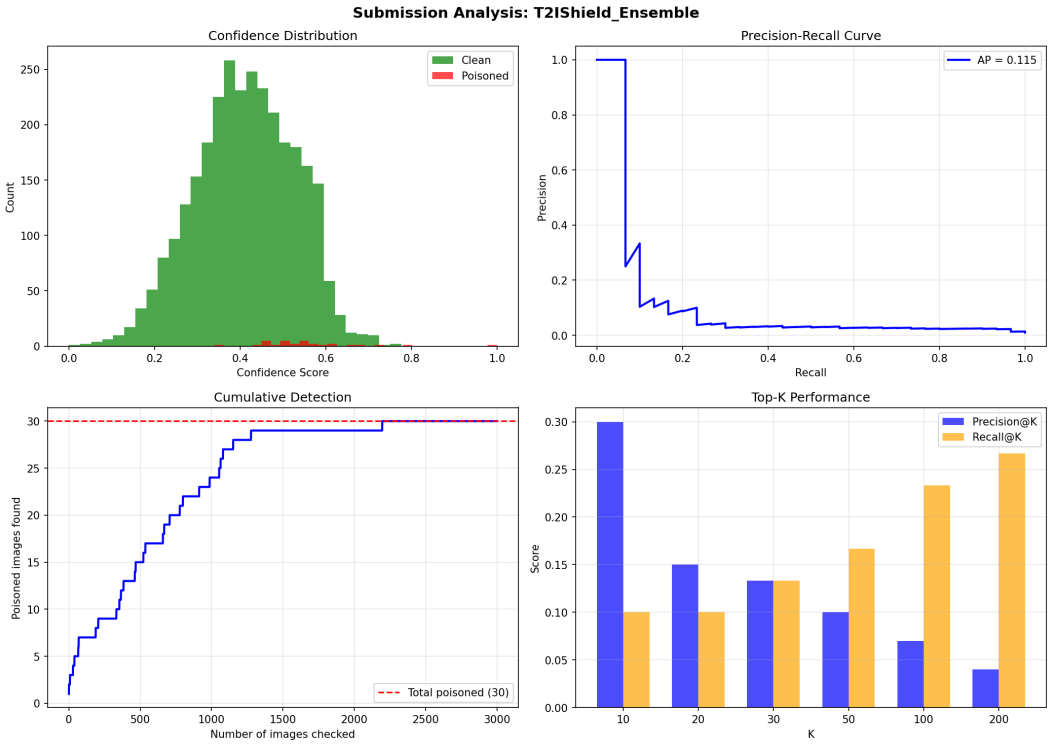

Figure 3: Ensemble submission analysis: (a) confidence score distributions for clean vs. poisoned images, (b) precision-recall curve (AP = 0.115), (c) cumulative poisoned images found vs. rank, (d) precision and recall at various K thresholds.

AP by 44% over the delta method alone. Nevertheless, P@30 remains low (0.133), reflecting the fundamental difficulty of this needle-in-haystack task at 1% poisoning rate.

4 Conclusion

We presented BackdoorHunter, showing that EvilEdit can inject a stealthy backdoor by editing 32 matrices in under one second, yet forensic weight analysis combined with attention-based FTT can partially recover the 30 implicated captions (AP = 0.115, AUROC = 0.810, P@30 = 0.133), an $8\times$ improvement over random but still reflecting the fundamental difficulty of this 1% needle-in-haystack detection task.

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